The Journal of Open Source Software

An Open-Source Reproduction and Enhancement of CheXNet for Chest X-ray Disease Classification

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DOI: 10.xxxxxx/draft

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Editor:

Submitted: 15 September 2025
Published: unpublished

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Summary

This project aimed to rigorously reproduce and extend CheXNet, a landmark deep learning model in medical imaging. Through our efforts, we demonstrated that meaningful improvements to the original architecture can be achieved by incorporating techniques developed since the paper's publication. Specifically, we found that the use of Focal Loss, the AdamW optimizer with weight decay, Color Jitter for data augmentation, ReduceLROnPlateau scheduling, and per-disease F1 threshold tuning substantially improved model stability and performance across all 14 thoracic disease classes. Our model, "DacNet," achieved a strong balance between interpretability and predictive power, reaching an average AUC of 0.85 and an F1 score of 0.39. These results highlight the potential for targeted enhancements to significantly improve performance on imbalanced and clinically relevant datasets like NIH ChestX-ray14. Beyond the technical contributions, this project underscores the importance of reproducibility in machine learning research. We were motivated not only to validate the claims of a high-impact study, but also to contribute openly to the scientific community. Our full codebase, including model configurations, evaluation tools, and visualization scripts, has been made publicly available in hopes of promoting transparency and enabling further development by others. Ultimately, this work contributes to both the biomedical and data science communities by showing how modern training strategies can elevate model performance in critical clinical applications. We hope our findings serve as a foundation for future work exploring deep learning in medical diagnostics and for building more accurate, interpretable, and equitable AI tools in healthcare.

Statement of need

Chest X-ray classification is a crucial task in medical image analysis, where deep learning models are trained to detect various thoracic diseases from radiographic scans. A landmark study in this field, known as CheXNet, introduced a 121-layer DenseNet convolutional neural network that reportedly outperformed radiologists in detecting pneumonia (Rajpurkar et al., 2017). Their work used the NIH ChestX-ray14 dataset, a publicly available dataset of over 100,000 frontal-view chest X-rays labeled with up to 14 disease classes (**?**{8099852). The success of CheXNet has inspired further research, as it represents a significant step toward using artificial intelligence to assist in clinical diagnosis, especially in regions where access to licensed radiologists is limited (Hwang et al., 2019). In the midst of a reproducibility crisis in academia, independent researchers must reproduce groundbreaking studies like this in order to help guide future research (Allison et al., 2016). In this project, we set out to replicate the original CheXNet model as closely as possible using the same NIH ChestX-ray14 dataset, evaluate and improve performance metrics such as AUC-ROC and F1 scores across all 14 disease classes, and explore whether newer deep learning techniques, particularly Vision Transformers (ViTs),

could offer performance improvements over traditional convolutional neural networks. All code for our models and evaluation pipeline is publicly available in our GitHub repository under an open-source MIT License.

While our primary goal was to replicate the original CheXNet study, we also recognize the importance of its successor, CheXNeXt, which validated a similar model against board-certified radiologists on a curated internal dataset (Rajpurkar et al., 2018). Although the test set used in CheXNeXt is not publicly available, its findings emphasize the clinical relevance of these models and reinforce the need for reproducible benchmarking in public datasets like NIH ChestX-ray14. Our key contributions are as follows: We performed a faithful replication of the CheXNet model, establishing a reproducible baseline using pretrained DenseNet-121 with standard training procedures. We proposed an improved model, DacNet, which incorporates Focal Loss, the AdamWoptimizer, and advanced image augmentations like Color Jitter. It achieved significantly higher F1 scores on rare classes compared to the baseline. We implemented per-class F1 threshold optimization to further boost classification accuracy, especially in multi-label settings. Unlike the original CheXNet study, which only reported an F1 score for pneumonia using a non-public expert-labeled subset, our study computes per-class F1 scores across all 14 diseases using a reproducible patient-wise split. This provides a more granular view of model strengths and limitations in multi-label medical image classification. We explored the use of transformer-based models (ViT) for X-ray classification, benchmarking their performance against CNN-based architectures. Finally, we developed a Streamlit web app hosted on Hugging Face that takes a chest X-ray input, returns disease predictions using DacNet, and overlays Grad-CAM heatmaps to visualize model attention.

Methods

We built on DenseNet-121 but replaced the BCE loss with Focal Loss (gamma=2, alpha=1) to address extreme class imbalance. We used the AdamW optimizer with weight decay, a learning rate of 0.00005, and a ReduceLROnPlateau scheduler. Augmentations included RandomResizedCrop(224), RandomHorizontalFlip, and ColorJitter. This model was trained using a patient-level split and achieved a test AUC of 0.85, a test loss of 0.04, and an average F1 score of 0.39. We believe the focal loss contributed significantly to reducing test loss and improving prediction confidence on minority classes. This model outperformed CheXNet in AUC for 9 out of 14 diseases.

Tables

Table 1: Performance vs older models and publications on Test AUC scores per disease.

Pathology	original CheXNet	Dacnet.py	vit_trans-former.py	repli-cate_chexnet.py
Atelectasis	0.8094	**0.817**	0.774	0.762
Cardiomegaly	0.9248	**0.932**	0.89	0.922
Consolidation	**0.7901**	0.783	0.789	0.746
Edema	0.8878	**0.896**	0.876	0.864
Effusion	0.8638	**0.905**	0.857	0.883
Emphysema	0.9371	**0.963**	0.828	0.85
Fibrosis	0.8047	**0.814**	0.772	0.766
Hernia	0.9164	**0.997**	0.872	0.925
Infiltration	**0.7345**	0.708	0.7	0.673
Mass	0.8676	**0.919**	0.783	0.824
Nodule	0.7802	**0.789**	0.673	0.646

Pathology	original CheXNet	Dacnet.py	vit_transformer.py	replicate_chexnet.py
Pleural Thickening	**0.8062**	0.801	0.766	0.756
Pneumonia	**0.768**	0.74	0.713	0.656
Pneumothorax	**0.8887**	0.875	0.821	0.827

Table 2: Average metrics across all diseases for each model.

Metric	DacNet	ViT Transformer	Replicate CheXNet
Loss	**0.0416**	0.1589	0.1661
AUC	**0.8527**	0.7940	0.7928
F1	**0.3861**	0.1114	0.0763

Table 3: F1 Score Comparison for Each Model (Per-disease).

Disease	DacNet	ViT Transformer	Replicate CheXNet
AVERAGE	**0.386**	0.111	0.076
Atelectasis	**0.421**	0.127	0.026
Cardiomegaly	**0.532**	0.264	0.423
Consolidation	**0.226**	0	0
Edema	**0.286**	0.004	0
Effusion	**0.623**	0.427	0.459
Emphysema	**0.516**	0.079	0
Fibrosis	**0.127**	0	0
Hernia	**0.750**	0	0
Infiltration	**0.395**	0.193	0.061
Mass	**0.477**	0.213	0.079
Nodule	**0.352**	0.041	0
Pleural Thickening	**0.258**	0	0
Pneumonia	**0.082**	0	0
Pneumothorax	**0.360**	0.211	0.021

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Figures

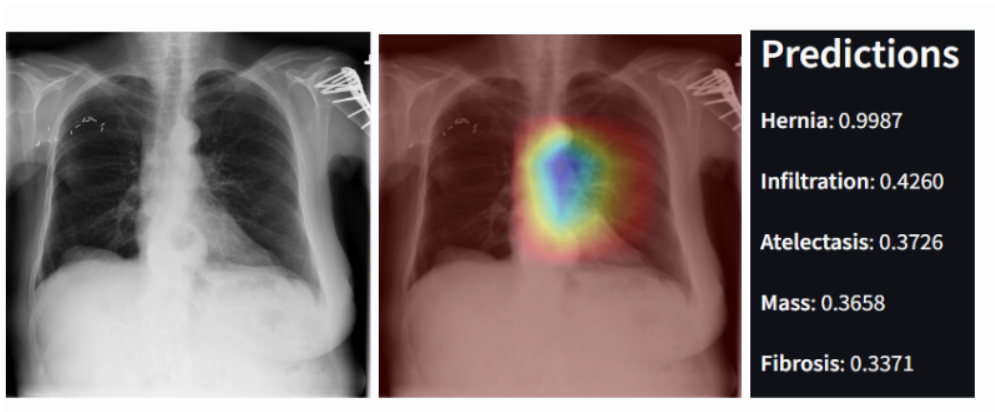

Figure 1: (Left) An example of an X-Ray from the NIH ChestX-ray14 with Hernia and Infiltration as its ground truth findings. (Center) Grad-CAM visualizations of the same X-Ray image. (Right) DACNet's predictions of the 5 most likely findings in the image.

Acknowledgements

We would like to thank Thomas Gardos, PhD for his mentorship and support.

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